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Global Facilities - Design & Construction Standard – Construction - Site Establishment

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on material and design specification that shall be followed and complied in the Design, Engineering and Construction of site establishment as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects with regards to provide site establishment as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.

2.0 Scope

- 2.1 This document for Site Establishment that covers all necessary related site facilities and site utilities installation to ensure full compliance of the contract requirement and required for full completion of the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all MICRON Facilities employees, contractors and vendors at all MICRON sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
GFCT	Global Facilities Construction Team
SME	Subject Matter Expert
QP	Qualified Person
EHS	Environmental, Health and Safety

4.0 Roles and Responsibilities

Team	Responsibilities
GFCT (Global Facilities Construction Project Management Teams)	• Manage and lead team on the activities the site establishment of any project, which includes the contractor's proposal review and acceptance • Ensure all execution are completed as per contract requirement, subsequently manage and support as appropriate with site project construction teams, contractors and vendors • Ensure related SMEs are included, as appropriate, to review the contractor's proposal to meet all related MICRON standard, local regulation, requirement, code and standard • Manage and track change management with site project construction team
Site Project Construction Teams	• Provide technical feedback on System Standard and contractor's proposal • Review and acceptance on contractor's proposal • Incorporate System Standard into project Design, Engineering and Construction • Manage and track change management • Document project deviation, proposal, expectation and any exclusion from System Standard • Ensure to capture in future version of System Standard
Site Project EHS Teams	• Supervision and ensure effective implementation of MICRON Design and Construction EHS Standards at site level and in compliance with local applicable Governmental Regulations. • Assist GFCT in review of contractor's proposal to meet or comply with all related MICRON Standards, local Regulations, requirements, local Codes and Standards. • Ensure that potential Contractors are appropriately briefed and communicated on MICRON Global EHS requirements and act as SME and technical expertise on EHS related design specifications.
Procurement Teams	• Review contractor's proposal on cost • Validate cost of change and change management • Ensure contractor's proposal must comply and to meet all related MICRON standard, local regulation, requirement, code and standard
GFTT Lead (Discipline Engineers)	• Develop and maintain System Standard Document • Manage and track change management with GFCT and site project construction team

Note:

- Writing highlighted in yellow - Only leave this comment intact and strike-off the rest in this segment if the document is meant for external Consultants/Contractors to comply with.
- Writing highlighted in grey - Requirements stated for internal use only. Not to be shown to 3rd parties outside MICRON.

5.0 Overview

5.1 Site Facilities

5.1.1 Control

5.1.2 Access

5.1.3 Support Facilities

5.2 Site Utilities

5.2.1 Temporary Utilities

6.0 Performance Specification

- 6.1 The Site Establishment comprises site facilities and site utilities shall be designed and installed to achieve the operational performance specification as outlined below. See **Table 1**.

Table 1: Operational Performance Specification for Site Establishment

Parameter	Specification Requirement
Control	Site Preparation, Barricade, Security, Environment Protection, Control & Maintenance
Access	Construction vehicle, Carpark & personnel access control, ingress & Egress
Support Utilities	Construction Office, Toilet, Rest & Laydown Area, Canteen & Dormitory
Temporary Utilities	Power, Water, Telecom & IT

7.0 Site Establishment Requirement

7.1 Materials Specification

All site establishment material specification for any green and/or brown field construction project shall comply to local authority regulation and standard requirement as well as full compliance to any applicable MICRON standard as stipulated below:

7.1.1 Control

7.1.1.1 **Security Guardforce deployment:** Certified local service provider.

7.1.1.2 **Guard House:** Hardcore base, foundation & concrete slab finish, container cabin guardhouse, table & chair.

7.1.1.3 **Worker & Site Personnel:** Turnstile gantry with facial recognition (where it is permissible by the local jurisdiction) & card access system, CCTV with 4K camera resolution and infra-red night vision capabilities. Cloud-based access shall be made available for multiple authorized users via secured internet interface.

External wide-angle CCTV shall be mount on pre-determined stationary high vantage point at 2 locations (minimum) to oversee the entire construction site, site office, material laydown area and holding area for work progress monitoring. Continuous time-lapse videos captured by such CCTs shall be made available for Micron's Cloud-base remoting access 24/7, from site occupation by the GC to completion. In addition, critical construction areas inside the buildings shall also be identified by the GC to be equipped with similar time-lapse CCTVs cameras at suitable phase/timing of construction. GC shall submit the proposed locations and phase/timing for the installation of these devices to Micron for review and acceptance.

7.1.1.4 **Surrounding site hoarding and gate:** Foundation footing, mild steel frame with corrugated metal sheet or aluminum panels, swing & sliding door & manual barrier control.

7.1.1.5 **Earth control measure & silt control:** Silt sediment concept in silt sump.

7.1.1.6 **Drain & sump:** Precast concrete drain and sump or equivalent with piles.

7.1.1.7 Environmental Protection & Control:

- i. **Environmental pollution prevention and control:** Meet local authority Act, regulation & standard requirement **such as Noise & dust barrier specification, contamination & hazardous material control requirement & parameter, etc.**
- ii. **Environmental sustainability:** Meet local authority Act, regulation & standard requirement **such as Recycle & green environment requirement & parameter, etc.**
- iii. **Pest Control:** Certified local service provider.

7.1.1.8 Housekeeping: General workers for housekeeping depend on project scale.

7.1.2 Access

7.1.2.1 Construction ingress, & egress: Hardcore base, concrete foundation & base.

7.1.2.2 Construction vehicle access & parking: Levelled hardcore base with asphalt finish.

7.1.2.3 Construction heavy equipment access & parking: levelled hardcore base with asphalt finish.

7.1.2.4 Wash bay: Hardcore base, concrete foundation/base & sump include steel structure wheel washer, water pump & sediment tank.

NOTE: Contractor is to meet minimum standards stipulated herein by MICRON. Contractor shall perform any necessary upgrade above and beyond the stipulated standard to comply to local code or to accommodate site construction requirements without imposing additional cost to MICRON (Employer) whenever applicable.

7.1.3 Support Facilities

7.1.3.1 Construction office with toilet: Hardcore base, foundation & concrete slab finishes, toilet partition, WC, urinal, basin and pantries, proprietary modular structural frame, external metal insulated wall panel, internal drywall with wall, floor & ceiling finishes, tables, chairs, phone, printer, photocopier & necessary stationeries, LAN & Wi-Fi connectivity.

- 7.1.3.2 **Construction toilet:** Hardcore base, foundation & concrete slab finishes, toilet partition, WC, urinal & basin, prefab containerized toilet housing with fully fitted sanitary fixtures or equivalent is preferred. Adequate ventilation system to be provided. Appropriate air-conditioning (heating) shall be provided in region with sub-zero winter conditions.
- 7.1.3.3 **Construction canteen** (Optional, depend on project scale)
- 7.1.3.4 **Construction rest area:** Levelled hardcore base with asphalt finish, shelter canopy with seating required. Water points shall be installed.
- 7.1.3.5 **Construction assembly & lay down area:** Levelled hardcore base with asphalt finish, steel frame and canvass tent, any flammable or hazardous materials storage on site shall be segregated from other construction materials with appropriate safety measures implemented.
- 7.1.3.6 **Temporary Construction Carpark:** Levelled hardcore base with asphalt finish.
- 7.1.3.7 **Workers' accommodation:** Proposed to limit the occupancy in each sleeping quarter to 6 pax or less with provision of full-height storage cabinet for individuals. No open-flame cooking facilities in common/shared kitchens. (Optional, depend on project scale)

7.1.4 Temporary Utilities

- 7.1.4.1 **Construction Power:** Incoming power supply from local electrical service provider, cabling, panel & distribution (genset include).
- 7.1.4.2 **Construction Water:** PVC piping tap from local authority water department supply.
- 7.1.4.3 **Construction IT:** IT for MICRON by MICRON vendor, IT for contractor by local service provider.
- 7.1.4.4 **Construction Lighting:** Power efficient LEDs.

7.2 Design Specification

All site establishment design specification for any green and/or brown field construction project shall comply to local authority regulation and standard requirement with necessary QP endorsement, required authority submission and approval as well as full compliance to any applicable MICRON standard as stipulated below:

7.2.1 Control

7.2.1.1 **Security Guard deployment:** 24/7 Security manning operation.

7.2.1.2 **Guard House:** Local contractor standard design requirement with QP endorsement, provision of ventilation, lighting & power, air-conditioning (heating) required for region with sub-zero winter condition to ensure achieving reasonable thermal comfort level for prolong occupancy under winter conditions. Basic intercom requirements needed for communication to site office for reporting.

7.2.1.3 **Worker & Site Personnel:** Anti-tailgate turnstile/mantrap system with integrated facial recognition (where it is permissible by the local jurisdiction) & access card control. Additional non-contact temperature scanner to be provided for the COVID pandemic period or any other similar situation in future. Cloud-based CCTV. Access shall be made available for multiple authorized users via secured internet interface. External wide-angle CCTV shall be mount on pre-determined stationary high vantage point at 2 locations (minimum) to oversee the entire construction site, site office, material laydown area and holding area for work progress monitoring. Continuous time-lapse videos captured by such CCTs shall be made available for Micron's Cloud-base remoting access 24/7, from site occupation by the GC to completion. In addition, critical construction areas inside the buildings shall also be identified by the GC to be equipped with similar time-lapse CCTVs cameras at suitable phase/timing of construction. GC shall submit the proposed locations and phase/timing for these devices installation to Micron for review and acceptance. Critical construction area inside the buildings shall be equipped with panoramic type camera and 90%~100% coverage for enhancement of safety and security.

- 7.2.1.4 **Surrounding site hoarding & gate:** Hoarding shall have adequate structure integrity & foundation support to withstand lateral wind load based on local climatic condition for personnel safety, local contractor standard design requirement with QP endorsement. Hoarding minimum height of at least 1.8m, with high-resolution, infra-red capable CCTVs for 24/7 monitoring. Sound-damping insulated hoarding panel shall be provided where local code requires.
- 7.2.1.5 **Earth control measure & silt control:** Local contractor & local authority design requirement with QP endorsement.
- 7.2.1.6 **Drain & sump:** All drains (precast or in situ), sumps and manhole shall be supported by concrete piles or equivalent against the effect of ground subsidence over time, local contractor & local authority design requirement with QP endorsement.
- 7.2.1.7 **Environmental Protection & Control:** MICRON's minimum Environmental sustainability Requirements:
- Construction material recycle collection point shall be provided within the site.
 - Temporary construction access road shall be construction with minimum 20% recycle materials.
 - Construction vehicle washing bay shall be equipped with auto sensor wheel-washers. Site should be fitted with sediment collection tanks (no discharge or public drain) from vehicle washing bays.
- 7.2.1.8 **Pest Control:** Pest control plan required to ensure site is pest free throughout the construction period and must compliance local authority requirement and local contractor pest control plan especially mosquitos, cockroach, rodent, etc.
- 7.2.1.9 **Environmental Pollution Prevention and Control:** Local contractor & local authority design requirement. SOIL and STORMWATER CONTAMINATION: stormwater collection and discharge system, isolation and appropriate containment of contamination sources (ex: generators, storage of oil/chemicals and waste...) to prevent soil and stormwater contamination. Spill control kits shall be available near storage, loading areas and transfer routes.
- WASTE COLLECTION and SEGREGATION: dedicated waste collection area, with segregation by type (hazardous and non-hazardous) and material to maximize the reuse/recycle/recovery of construction and office material (offsite disposal).
- NOISE CONTROL: limit day/night noise levels, identify noise sources (equipment/construction phase or activity) and implement mitigation measures, define monitoring plan.

Particular focus on potential sensitive receptors in the surrounding areas (hospital, schools, residential areas).

DUST CONTROL: define a plan and control measures to minimize the dust generation during excavation, backfilling and other activities.

7.2.1.10 **Environmental Sustainability:** Local authority design requirement.

- Establish a monitoring plan to keep track of key environmental indicators: energy and water consumption, total waste generated and waste reuse/recycle/recovery.
- Define a plan for the reduction of energy use, water use and waste generation.
- Use of recycled construction materials where available and appropriate.
- Environmental & recycle plan to meet MICRON standard, local practice & standard.

7.2.1.11 **Housekeeping:**

Housekeeping plan to meet MICRON standard, local practice & standard.

- All hazardous and/or combustible building materials stored on site to be housed in separate storage shed and be alerted with appropriate warning sign.

7.2.2 Access

7.2.2.1 **Construction ingress, & egress:** Local contractor standard design requirement with QP endorsement. As per local code and local best practice safety standard.

7.2.2.2 **Construction vehicle access & parking:** Local contractor standard design requirement with QP endorsement. As per local code and local best practice safety standard.

7.2.2.3 **Construction heavy equipment access & parking:** Local contractor standard design requirement with QP endorsement. As per local code and local best practice safety standard.

7.2.2.4 **Wash bay:** Local contractor standard & local authority design requirement with QP endorsement. Construction vehicle washing bay shall be equipped with auto sensor wheel-washers. Site should be fitted with sediment collection tanks (no discharge or public drain) from vehicle washing bays.

7.2.3 Support Facilities

- 7.2.3.1 **Construction office with toilet:** Construction office workstation, manager & meeting room, pantry & toilet concept (hand free sensor tap & soap dispensers system , hygiene control), data/internet connectivity capacity (Telephone, LAN points, WIFI) & requirement, stationery, copier, projector, white board & telecommunication to meet MICRON standard, local practice & standard. There shall be 1 number of handicapped accessible toilet cubicle (with wash basin) available for both male & female toilets. Toilets to be located on ground floor to facilitate handicapped access for construction site office personnel.
- 7.2.3.2 **Construction toilet:** Same as 7.1.3.2, local contractor standard design requirement with QP endorsement.
- 7.2.3.3 **Construction canteen** (Optional, depend on project requirement)
- 7.2.3.4 **Construction rest area:** Same as 7.1.3.4, local contractor standard & local practice design requirement with QP endorsement, no requirement yet on COVID compliance.
- 7.2.3.5 **Construction assembly & lay down area:** Local contractor standard & local practice design requirement with QP endorsement.
- 7.2.3.6 **Temporary Construction Carpark:** Same as 7.1.3.7, local contractor standard design requirement with QP endorsement.
- 7.2.3.7 **Workers' accommodation:** Approved service provider to meet MICRON requirement.

7.2.4 Temporary Utilities

- 7.2.4.1 **Construction Power:** Local contractor standard design requirement.
- 7.2.4.2 **Construction Water:** Local contractor standard design requirement.
- 7.2.4.3 **Construction IT:** Local contractor & local service provider specification requirement (only MICRON specification for MICRON IT).
- 7.2.4.4 **Construction Lighting:** Minimum Lux level for general activities on site, security surveillance and for the respective trades' safe working environment requirements shall be provided to meet OSHA standards or local code whichever is more stringent.

7.3 Standard Detailing & Drawing

All site establishment standard detailing & drawing for any green and/or brown field construction project shall be subjected to MICRON approval and full compliance to the material & design specification stipulated in 7.1 & 7.2.

7.3.1 Control

7.3.1.1 **Guard House:** Local contractor standard guard house detail.

7.3.1.2 **Worker & Site Personnel:** Turnstile, facial recognition (where it is permissible by the local jurisdiction) & temperature scanner to manufacturer's detail, done to meet MICRON's standards or local code requirements, whichever is more stringent. CCTV with 4K camera resolution and infra-red night vision capabilities. Cloud-based access shall be made available for multiple authorized users via secured internet interface. Suitable model shall be recommended by the GC's appointed security consultant.

7.3.1.3 **Surrounding site hoarding & gate:** Local contractor standard gate & hoarding detail and height.

7.3.1.4 **Earth control measure & silt control:** Local manufacturer standard & detail for silt treatment & filtration system.

7.3.1.5 **Drain & sump:** Local contractor standard drain & sump detail, size, width & depth.

7.3.1.6 **Environmental Protection & Control:** Local contractor standard pollution control, recycle & green proposal & system detail.

For example,

Taiwan

a. Management Regulations for Construction Project Air Pollution Control Facilities

b. Water Pollution Control Act

c. Amendment to Noise Control Standards

Singapore

a. Earth Control Measures at Construction Site - Control of silt discharge into storm water drainage system

b. Noise Control Measures & Guidelines on boundary of Construction Site - Max. noise limit and duration to residential development near work sites.

c. Water and Air Contamination Control Measures at Construction Site - Air and water quality sampling near work sites.

7.3.1.7 Environmental Sustainability: Local contractor standard pollution control, recycle & green proposal & system detail. For example, Waste Disposal Act, etc.

7.3.2 Access

7.3.2.1 Construction vehicle access & parking: Local contractor standard access detail & width.

7.3.2.2 Construction heavy equipment access & parking: Local contractor standard access detail & width.

7.3.2.3 Construction ingress, & egress: Local contractor standard access detail & width.

7.3.2.4 Wash bay: Local contractor standard wash bay detail, size, width & depth, manufacturer system & detail for recycle, silt treatment & filtration system.

7.3.3 Support Utilities

7.3.3.1 Construction office with toilet: Local contractor proposal detail, size, width and number of stories required.

7.3.3.2 Construction toilet: Local contractor proposal detail, size, width and number of stories required.

7.3.3.3 Construction canteen (Optional, depend on project requirement)

7.3.3.4 Construction rest area: Local contractor standard rest area detail.

7.3.3.5 Construction assembly & lay down area: Local contractor standard assembly & lay down detail.

7.3.3.6 Temporary Construction Carpark: Local contractor standard carpark detail.

8.0 Plan & Layout Proposal

All site establishment plan & layout proposal for any green and/or brown field construction project shall be subjected to MICRON approval and full compliance to the material & design specification stipulated and standard detailing drawing in 7.1, 7.2 & 7.3.

9.0 Document Control

9.1 Approval

FCG_ENG_CONST_APPR

9.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_NOTIFY
GLOBAL_FAC_CONST_SOP
GLOBAL_SECURITY_TEAM
GLOBAL_EHS_SAFETYHEALTH
GP_INDIRECT_FACILITIES
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
WW_FACILITIES_MANAGERS_APPR

9.3 Final Viewer Access

All MICRON Team Members at all Sites

9.4 Retention

Adherence to the requirements specified in
[**Global Facilities - Document Control Procedures**](#)

9.5 Review

Biennial

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10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	01/27/21
	Section 7	7.1.1.7. Provided examples for Environmental pollution prevention and control and for Environmental sustainability 7.1.1.3 Addition on ext wide camera requirement, time-lapse videos captured and panoramic type camara. 7.2.1.9 Added section on Environmental Pollution Prevention and Control 7.2.1.10 Added "Local authority design requirement" 7.3.1.6 Added section on Environmental Protection & Control 7.3.1.7 Added section on Environmental Sustainability	11/10/21
	Section 9	9.2 Removed FCG_F2_FAC_NOTIFY	
	Appendix 1	- Added remarks for Performance bond - Added a row for MICRON requirements on EHS - Added remarks for MICRON requirement on testing & commission, start-up of system	
	Appendix 2	- Added information for Worker & Site Personnel under Material Specification and Design Specification - Added a row for Environmental pollution prevention and control - Added a row for Environmental Sustainability	

11.0 Appendix:

11.1 Reference Specifications

No.	Specifications
02201	Structural earthwork
02220	Trenching _ Backfilling
02510	Asphalt concrete paving
02520	Concrete sitework
02530	Pavement markings
02600	Water System
02720	Storm Water Systems
02740	Sewer System
02800	Fences Gates
05400	Cold Formed Metal Framing
05500	Metal fabrications
07600	Flashing and Sheet Metal
10800	Toilet Accessories
033000	Cast-in-place concrete
033543	Polished Concrete Finishing
051200	Structural Steel Framing

11.2 Reference Criteria

11.2.1 General Condition & Document

- Refer to Excel Sheet – [Appendix 1](#)

11.2.2 Site Establishment Standard

- Refer to Excel Sheet – [Appendix 2](#)

11.2.3 EHS Safety Guidebook

- Refer to the [Micron Construction Safety Guidebook](#)

Appendix 1: General Condition & Document

L4 WBS	L5 WBS Element	Task & Activities	Material Specification	Design Specification	Standard Detailing & Drawing	Remark
General	Conditions of contract	Dilapidation/ pre-condition survey	Certified local service provider	Local contractor survey requirement to meet local practice & standard with QP endorsement	Local contractor available pre-condition survey detail	Documentation to be submitted by licensed/registered surveyor with endorsement as per local code requirements
			Local contractor standard survey requirement & parameter			
		Cable detection	Certified local service provider	Local contractor Cable detection requirement & plan with license cable detector to meet local authority requirement & standard with endorsement	Local contractor available cable detection detail	Beside cable detection, licensed surveyors and\ or relevant P.E.s & QP will also be required to provide: 1. Geotechnical Survey Report 2. Soil Investigation Report 3. Environmental Impact Assessment (EIA) Report 4. Soil Contaminant Assessment Report 5. Traffic Impact Assessment 6. Topographic Survey Report 7. Underground Services Detection Survey Report Necessary survey reports shall be provided by the appointed General Contractor (GC). If these reports were given to the GC by MICRON upon award, the information shall be served as reference only. All reports shall be verified and owned by the GC.
			Cable detection, trial pit, services drawings as built			
		Performance bond	Approved service provider	Local practice standard performance bond to meet MICRON requirement & parameter	Local practice performance bond specimen	GC to refer MICRON Procurement for any specific Performance Bonds and "MICRON Master File Checklist for Construction", if necessary and required.
			MICRON standard bond requirement & parameter			
		Contractor All Risks, Workers Compensation insurance	Approved service provider	Local practice standard & local authority insurance coverage to meet MICRON requirement	Local practice insurance specimen	GC to refer MICRON Procurement for any specific Insurance Coverage, if necessary and required.
			MICRON standard & local authority standard insurance coverage requirement			
		Protection of works	Local contractor & MICRON construction standard requirement	Local contractor protection of works plan & design to meet MICRON standard, local practice & standard	Local contractor standard protection detail	
		Inspection of material & work	MICRON standard & construction standard requirement	Local contractor inspection to meet local authority requirement & standard	Local contractor & local consultant inspection standard	
		Testing of material	MICRON standard & construction standard requirement	Local contractor testing to meet local authority requirement & standard	Local contractor & local consultant inspection standard	
		FAT of part, equipment & material	MICRON standard & construction standard requirement	Local contractor FAT to meet MICRON standard, local practice & standard	Local contractor and manufacturer system & detail	
		MICRON requirements on EHS	MICRON standard & construction standard requirement Contractor EHS Requirements, refer to "MICRON Global EHS Standard Guidebook", "RBA (Responsible Business Alliance)" and "MICRON Construction / Design Construction Contractor EHS Minimum Performance Requirements", etc.	Local contractor EHS to meet MICRON standard, local practice & standard. Local contractors provide "Project Construction Protection Plan" which shall be reviewed and approved by MICRON EHS team prior construction work start.		Besides meeting local code on EHS, MICRON team established global EHS standard over & above local standards. E.g. Construction PPE standard vs local practices. GC to refer MICRON EHS for compliance requirement (MICRON Safety Guidebook provided)
		MICRON requirements on RBA	MICRON standard & construction standard requirement	RBA to meet MICRON standard		GC to refer MICRON Procurement & EHS for any RBA compliance requirement
		MICRON requirements on quality control	MICRON standard & construction standard requirement	Local contractor QAQC to meet MICRON standard, local practice & standard		
Standard Documentation	Drawings	Construction drawing, as built drawings	Local contractor & MICRON construction standard requirement	Local contractor drawing format to meet MICRON standard, local practice & standard		As-built drawings shall be endorsed by licensed surveyor and QP independent, whichever applicable, from the GC's team as a minimum. Methodology of survey may include 3D laser scanning, GPS fixing if need be and necessary Opportunity to integrate as-built survey info with LOD400/500 integrated BIM data.
	Manual	Operation & maintenance manual	Local contractor & MICRON construction standard requirement	Local contractor O&M format to meet MICRON standard, local practice & standard		
		Catalogue & test report	Local contractor & MICRON construction standard requirement	Local contractor catalogue & test report format to meet MICRON standard, local practice & standard		
	Material & parts	Spares	Local contractor & MICRON construction standard requirement	Local contractor & practice spares list to meet MICRON standard	Local practice spare list detail	
	Report	Progress report, scheduling	Local contractor standard requirement (MICRON global construction project reporting template)	Local contractor various report format to meet MICRON standard	Local practice report detail	
	Standard	Forms	Local contractor & MICRON construction standard requirement	Local contractor various forms format to meet MICRON standard, local practice & standard	Local practice form detail	
		Labelling	Local contractor & MICRON construction standard requirement	Local contractor labelling format to meet MICRON standard, local practice & standard	Local practice labelling detail	

Appendix 2: Site Establishment Standard

L4 WBS	L5 WBS Element	Task & Activities	Material Specification	Design Specification	Standard Detailing &	Plan & Layout
Site Facilities	Control	Security guardforce deployment	Certified local service provider	24/7 security manning operation		
		Guard house	Hardcore base, foundation & concrete slab finish, container cabin guardhouse, table & chair	Local contractor standard design requirement with QP endorsement, provision of ventilation, lighting & power, air-conditioning (heating) required for region with sub-zero winter condition to ensure achieving reasonable thermal confort level for prolong occpancy under winter conditions. Basic intercom	Local contractor standard guard house detail	Local contractor site plan & layout proposal
		Worker & site personnel	Turnstile gantry with facial recognition (where it is permissible by the local jurisdiction) & card access system	Anti-tailgate turnstile/mantrap system with integrated thermal facial recognition system (where it is permissible by the local jurisdiction) which is equipped HD camera with infrared night vision & WIFI and sufficient device database, access card control. Additional non-contact temperature scanner to be provided for the COVID pandemic period or any other similar situation in future Scorecard points will be allocated for complying bidders (GCs) in the tender process for construction methodology that provides earlier erection of Fast Deployment Containerized Gantry Enclosure serving as construction site access control.	Turnstile, facial recognition (where it is permissible by the local jurisdiction) & temperature scanner to manufacturer's detail, done to meet MICRON's standards or local code requirements, whichever is more stringent	Local contractor site access, security plan & layout proposal to meet MICRON requirement and specification
			CCTV with 4K camera resolution and infra-red night vision capabilities. Cloud-based access shall be made available for multiple authorized users via secured internet interface External wide-angle CCTV shall be mount on pre-determined stationary high vintage point at 2 locations (minimum) to oversee the entire construction site, site office, material laydown area and holding area for work progress monitoring. Continuous time-lapse videos captured by such CCTs shall be made available for Micron's Cloud-base remoting access 24/7, from site occupation by the GC to completion. In addition, critical construction areas inside the buildings shall also be identified by the GC to be equipped with similar time-lapse CCTVs cameras at suitable phase/timing of construction. GC shall submit the proposed locations and phase/timing for these devices installation to Micron for review and acceptance	Cloud-based CCTV. Access shall be made available for multiple authorized users via secured internet interface External wide-angle CCTV shall be mount on pre-determined stationary high vintage point at 2 locations (minimum) to oversee the entire construction site, site office, material laydown area and holding area for work progress monitoring. Continuous time-lapse videos captured by such CCTs shall be made available for Micron's Cloud-base remoting access 24/7, from site occupation by the GC to completion. In addition, critical construction areas inside the buildings shall also be identified by the GC to be equipped with similar time-lapse CCTVs cameras at suitable phase/timing of construction. GC shall submit the proposed locations and phase/timing for these devices installation to Micron for review and acceptance. Critical construction area inside the buildings shall be equipped with panoramic type camara and 90%~100% coverage for enhancement of safety and security	CCTV with 4K camera resolution and infra-red night vision capabilities. Cloud-based access shall be made available for multiple authorized users via secured internet interface. Suitable model shall be recommended by the GC's appointed security consultant	
		Surrounding site hoarding & gate	Foundation footing, mild steel frame with corrugated metal sheet or aluminum panels, sliding door & manual barrier control	Hoarding shall have adequated structure integrity & foundation support to withstand lateral wind load based on local climateic condition for personnel safety, local contractor standard design requirement with QP endorsement Hoarding minimum height of at least 1.8m, with high-res, infra-red capable CCTVs for 24/7 monitoring. Sound-damping	Local contractor standard gate & hoarding detail, height	Local contractor site plan & layout proposal

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		Earth control measure & silt	Silt sediment concept in silt sump	Local contractor & local authority design requirement with QP endorsement	Local manufacturer standard & detail for silt treatment & filtration	Local contractor site plan & layout proposal
		Drain & sump	Precast concrete drain & sump or equivalent with piles	All drains, sumps and manhole shall be supported by concrete piles or equivalent against the effect of ground subsidence over time, local contractor & local authority design requirement with	Local contractor standard drain & sump detail, size width & depth	Local contractor site plan & layout proposal
		Pest control	Certified local service provider	Pest control plan required to ensure site is pest free throughout the construction period and must compliance local authority requirement and local contractor pest control plan especially		
		Environmental pollution prevention and control	Local authority Act, regulation & standard requirement such as Noise & dust barrier specification, contamination & hazardous material control requirement & parameter, etc	Local contractor & local authority design requirement SOIL and STORMWATER CONTAMINATION: stormwater collection and discharge system, isolation and appropriate containment of contamination sources (ex: generators, storage of oil/chemicals and waste...) to prevent soil and stormwater contamination. Spill control kits shall be available near storage, loading areas and transfer routes WASTE COLLECTION and SEGREGATION: dedicated waste collection area, with segregation by type (hazardous and non-hazardous) and material to maximize the reuse/recycle/recovery of construction and office material (offsite disposal) NOISE CONTROL: limit day/night noise levels, identify noise sources (equipment/construction phase or activity) and implement mitigation measures, define monitoring plan. Particular focus on potential sensitive receptors in the surrounding areas (hospital, schools, residential areas,..)	Local contractor standard pollution control system & detail: For example, Taiwan a. Management Regulations for Construction Project Air Pollution Control Facilities b. Water Pollution Control Act c. Amendment to Noise Control Standards Singapore 1. Earth Control Measures at Construction Site - Control of silt discharge into storm water drainage system 2. Noise Control Measures &	
		Environmental Sustainability	Local authority Act, regulation & standard requirement such as Recycle & green environment requirement & parameter, etc	Establish a monitoring plan to keep track of key environmental indicators: energy and water consumption, total waste generated and waste reuse/recycle/recovery Define a plan for the reduction of energy use, water use and waste generation Use of recycled construction materials where available and appropriate Environmental & recycle plan to meet MICRON standard, local	Local contractor standard recycle & green proposal & detail: For example, Waste Disposal Act, etc	
		Housekeeping	General workers for housekeeping depend on project scale	Housekeeping plan to meet MICRON standard, local practice & standard. All hazardous and/or combustible building materials stored on site to be housed in separate storage shed and be alerted with appropriate warning sign	Local contractor housekeeping proposal & detail	

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L4 WBS	L5 WBS Element	Task & Activities	Material Specification	Design Specification	Standard Detailing & Drawing	Plan & Layout
	Access	Construction vehicle access & parking	Levelled hardcore base with asphalt finish	Local contractor standard design requirement with QP endorsement As per local code and local best practice safety standard	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Construction heavy equipment access & parking	Levelled hardcore base with asphalt finish	Local contractor standard design requirement with QP endorsement As per local code and local best practice safety standard	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Construction ingress, & egress	Hardcore base, concrete foundation & base	Local contractor standard design requirement with QP endorsement As per local code and local best practice safety standard	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Wash bay	Hardcore base, concrete foundation/ base & sump include steel structure wheel washer, water pump & sediment tank	Local contractor standard & local authority design requirement with QP endorsement. Construction vehicle washing bay shall be equipped with auto sensor wheel-washers. Site should be fitted with sediment collection tanks (no discharge to public drain) from vehicle washing bays	Local contractor standard wash bay detail, size, width & depth, manufacturer system & detail for recycle, silt treatment & filtration system	Local contractor site plan & layout proposal
Site Facilities	Support facilities	Construction office with toilet	Hardcore base, foundation & concrete slab finishes, toilet partition, WC, urinal, basin & pantries	Local contractor standard design requirement with QP endorsement, provision of air conditioning, lighting & power	Local contractor standard office detail, size, width and number of storeys required	Local contractor site plan & layout proposal
			Proprietary modular structural frame, external metal insulated wall panel, internal drywall with wall, floor & ceiling finishes, tables, chairs, phone, printer, photocopier & necessary stationeries, LAN & Wi-Fi connectivity	Construction office workstation, manager & meeting room (social distancing), pantry & toilet concept (hand free sensor tap & soap dispensers system, hygiene control, social distancing), data/internet connectivity capacity (Telephone, LAN points, WIFI) & requirement, stationery, copier, projector, white board & telecommunication to meet MICRON standard, local practice & standard. There shall be 1 no. of handicapped accessible toilet cubicle (with wash basin) available for both male & female toilets. Toilets to be located on ground floor to facilitate handicapped access for construction site office personnel	Local contractor proposal detail	Local contractor proposal on cubicle, office, meeting room, pantry, toilet area, stationery, reception, document room
			Toilet interior, wall floor & ceiling finishes			
			Phone, printer, photocopier & necessary stationeries			

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		Construction heavy equipment access & parking	Levelled hardcore base with asphalt finish	Local contractor standard design requirement with QP endorsement As per local code and local best practice safety standard	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Construction ingress, & egress	Hardcore base, concrete foundation & base	Local contractor standard design requirement with QP endorsement As per local code and local best practice safety standard	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Wash bay	Hardcore base, concrete foundation/ base & sump include steel structure wheel washer, water pump & sediment tank	Local contractor standard & local authority design requirement with QP endorsement. Construction vehicle washing bay shall be equipped with auto sensor wheel-washers. Site should be fitted with sediment collection tanks (no discharge to public drain) from vehicle washing bays	Local contractor standard wash bay detail, size, width & depth, manufacturer system & detail for recycle, silt treatment & filtration system	Local contractor site plan & layout proposal
Site Facilities	Support facilities	Construction office with toilet	Hardcore base, foundation & concrete slab finishes, toilet partition, WC, urinal, basin & pantries	Local contractor standard design requirement with QP endorsement, provision of air conditioning, lighting & power	Local contractor standard office detail, size, width and number of storeys required	Local contractor site plan & layout proposal
			Proprietary modular structural frame, external metal insulated wall panel, internal drywall with wall, floor & ceiling finishes, tables, chairs, phone, printer, photocopier & necessary stationeries, LAN & Wi-Fi connectivity	Construction office workstation, manager & meeting room (social distancing), pantry & toilet concept (hand free sensor tap & soap dispensers system, hygiene control, social distancing), data/internet connectivity capacity (Telephone, LAN points, WIFI) & requirement, stationery, copier, projector, white board & telecommunication to meet MICRON standard, local practice & standard. There shall be 1 no. of handicapped accessible toilet cubicle (with wash basin) available for both male & female toilets. Toilets to be located on ground floor to facilitate handicapped access for construction site office personnel	Local contractor proposal detail	Local contractor proposal on cubicle, office, meeting room, pantry, toilet area, stationery, reception, document room
			Toilet interior, wall floor & ceiling finishes			
			Phone, printer, photocopier & necessary stationeries			

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L4 WBS	L5 WBS Element	Task & Activities	Material Specification	Design Specification	Standard Detailing & Drawing	Plan & Layout
		Construction toilet	Hardcore base, fondation & concrete slab finishes, prefab containerized toilet housing with fully fitted sanitary fixtures or equivalent is preferred. Adequate ventilation to be provided. Appropriate active heating system for region with sub-zero winter conditions.	Local contractor standard design requirement with QP endorsement		Local contractor site plan & layout proposal
			Toilet partition, WC, urinal & basin	No requirement yet on COVID compliance		
		Construction canteen				Local contractor site plan & layout proposal
		Construction rest area	Levelled hardcore base with asphalt finish	Local contractor standard & local practice design requirement with QP endorsement	Local contractor standard rest area detail	Local contractor site plan & layout proposal
			Steel frame and canvass tent	No requirement yet on COVID compliance		
		Construction assembly & laydown area	Assembly & laydown area levelled hardcore base with concrete finish	Local contractor standard & local practice design requirement with QP endorsement	Local contractor standard assembly & laydown detail	Local contractor site plan & layout proposal
			Steel frame and canvass tent	Local contractor standard & local practice design requirement with QP endorsement		
		Temporary construction carpark	Levelled hardcore base with asphalt finish	Local contractor standard design requirement with QP endorsement	Local contractor standard access detail & width	Local contractor site plan & layout proposal
		Workers' accommodation		Approved service provider to meet MICRON requirement		
Site Utilities	Temporary utilities	Construction power	Incoming power supply from local electrical service provider, cabling, panel & distrubution (genset include)	Local contractor standard design requirement		Local contractor site plan & layout proposal
		Construction water	PVC piping tap from local authority water department supply	Local contractor standard design requirement		Local contractor site plan & layout proposal
		Construction IT	IT for MICRON by Micorn vendor	Local contractor & local service provider specification requirement (only MICRON specification for MICRON IT)		
			IT for contractor by local service provider			
		Construction Lighting	power efficient LEDs	Minimum Lux level for general activities on site, security surveillance and for the respective trades' safe working environment requirements shall be provided to meet OSHA standards or local code whichever is more stringent		